Weekly Literature Review

Period: Nov 04 – Nov 11, 2026

This Week in Brief

The biggest news is the first Lp(a)-lowering outcomes trial. Pelacarsen suppressed Lp(a) deeply but did not reduce cardiovascular events (HR 1.02), and ziltivekimab also failed on MACE despite potent IL-6 neutralization, unsettling two of cardiology's most anticipated new targets. Established tools fared better: left bundle-branch pacing beat RV pacing on outcomes in AV block. For transplant practice, dapagliflozin modestly improved eGFR slope in stable heart transplant recipients without excess infections, supporting its use when kidney function is declining. Elsewhere, randomized AI trials in AI-ECG-guided AF screening and a maternal health chatbot showed fragile or no benefit.

Cardiology

A landmark week split between disappointments for new targets and gains for established tools. Lp(a) lowering with pelacarsen and IL-6 blockade with ziltivekimab both failed on hard outcomes despite large biomarker effects. Meanwhile, conduction-system pacing beat RV pacing on outcomes, the two Brazilian OPTIMAL trials split on intensive blood-pressure targets, and SGLT2 inhibition moved into heart transplant recipients.

Pelacarsen and Cardiovascular Outcomes in Patients with High Lipoprotein(a)

N Engl J Med · Cho L, Nicholls SJ, Frndak S, et al. · Placebo-controlled RCT, 8,350 patients · DOI

Area: Lipoprotein(a), secondary prevention

Design: Patients with established cardiovascular disease and Lp(a) ≥70 mg/dL received monthly subcutaneous pelacarsen 80 mg or placebo. Primary endpoint: CV death, nonfatal MI, nonfatal stroke, or urgent coronary revascularization. Median follow-up 4.7 years.

Results: Median baseline Lp(a) was 236 nmol/L; at 12 months it was 64.4 vs 233.4 nmol/L. The primary endpoint occurred in 12.3% vs 12.1% (HR 1.02; 95% CI 0.90 to 1.15; P=0.76). With baseline Lp(a) ≥90 mg/dL, rates were 12.3% vs 12.3% (HR 1.00; 95% CI 0.87 to 1.15). Serious adverse events occurred in 37.3% vs 41.1%.

Context: The first Lp(a)-lowering outcomes trial is flatly null, with an HR of 1.02 despite deep, sustained Lp(a) suppression and no signal in the highest-Lp(a) stratum. This undercuts the assumption, drawn from Mendelian randomization, that lowering Lp(a) pharmacologically will translate into fewer events over five years. Possible explanations include insufficient absolute reduction, the entry threshold and the time

horizon. The siRNA programs that achieve deeper suppression now carry much more uncertainty. Lp(a) remains a risk marker, but it no longer implies an imminent therapy.

Interleukin-6 Inhibition with Ziltivekimab for Atherosclerotic Disease

N Engl J Med · Ridker PM, Lincoff AM, Marx N, et al. · Double-blind RCT, 6,376 patients · DOI

Area: Anti-inflammatory therapy, ASCVD with CKD

Design: Adults with atherosclerotic disease, chronic kidney disease and hsCRP ≥2 mg/L received ziltivekimab 15 mg subcutaneously monthly or placebo. Primary endpoint: 3-point MACE. Median follow-up 33 months.

Results: At 24 months, placebo-adjusted median changes were -76 percentage points for free IL-6 and -64 for hsCRP. MACE occurred in 17.6% vs 17.7% (HR 0.99; 95% CI 0.88 to 1.11; P=0.84). Serious adverse events were 54.6% vs 54.8%, but serious infections were 20.4% vs 17.9%.

Context: Potent IL-6 neutralization had no effect on MACE (HR 0.99) in the population thought most likely to benefit: high residual inflammatory risk on top of CKD. The residual-inflammatory-risk paradigm built on CANTOS and the colchicine trials now looks much less secure, and hsCRP reduction appears not to be a reliable surrogate for event reduction. The numerically higher rate of serious infections adds cost without benefit. The ongoing ziltivekimab work in HFpEF must now stand on its own rationale, not on an atherosclerosis win.

Left Bundle-Branch Pacing versus Right Ventricular Pacing for Atrioventricular Block

N Engl J Med · Qian Z, Jiang L, Hou X, et al. · Open-label RCT, 702 patients · DOI

Area: Conduction-system pacing, AV block

Design: Patients with AV block were randomized to left bundle-branch pacing or right ventricular pacing. Primary composite: all-cause death, heart failure hospitalization, or pacing-induced cardiomyopathy. Median follow-up 35.1 months; stopped early at the prespecified interim analysis.

Results: In the modified intention-to-treat population of 685, the primary outcome occurred in 16 vs 35 patients. Four-year cumulative incidence was 8.5% vs 19.2% (HR 0.41; 95% CI 0.23 to 0.75; P=0.003). Serious adverse events unrelated to implantation were 11.8% vs 9.3% (P=0.32).

Context: This is the first randomized outcomes evidence that left bundle-branch pacing beats RV pacing in AV block, with the composite falling from 19.2% to 8.5% at four years. Until now, support for conduction-system pacing came from observational and echocardiographic data. The caveats are real: open-label design, early stopping that can inflate effect size, few events, and a composite that includes pacing-induced cardiomyopathy, a softer endpoint. Even so, it will push LBBAP toward default for patients expected to need substantial ventricular pacing, leaving lead durability and operator learning curves as the open questions.

Intensive Blood-Pressure Lowering in Patients with Type 2 Diabetes

N Engl J Med · Santo K, Martins E, Silva GS, et al. · Open-label RCT, 9,476 patients · DOI

Area: Hypertension in type 2 diabetes

Design: Brazilian patients with type 2 diabetes, elevated blood pressure and high cardiovascular risk were randomized to a systolic target of <120 vs <140 mm Hg. Primary composite: CV death, nonfatal MI, nonfatal stroke, or hospitalization for unstable angina or heart failure. Median follow-up 4.0 years.

Results: Mean achieved systolic pressure was 121.8 vs 135.4 mm Hg. The primary outcome occurred in 6.0% vs 6.7% (1.65 vs 1.83 events per 100 person-years; HR 0.90; 95% CI 0.77 to 1.06; P=0.20). Serious adverse events were 14.9% vs 14.6%.

Context: Targeting a systolic pressure below 120 in diabetes did not significantly reduce MACE, echoing ACCORD-BP and sitting in tension with the Chinese BPROAD trial, which reported benefit. The point estimate is compatible with a modest benefit, and event rates were low, so the trial may have been underpowered. The companion OPTIMAL-STROKE trial, run on the same protocol, was positive. Meta-analyses will likely show a small benefit of intensive control in diabetes, but this result argues against mandating <120 for patients with diabetes.

Dapagliflozin for Renal Protection in Heart Transplant Recipients: A Randomized Clinical Trial

Circulation · Holmen S, Manintveld O, Damman K, et al. · Double-blind RCT, 430 patients · DOI

Area: Heart transplantation, kidney protection

Design: Recipients at least one year post-transplant with eGFR ≥25, at six Northern European centers, received dapagliflozin 10 mg or placebo daily. Primary endpoint: eGFR slope from 2 weeks to 12 months.

Results: Baseline eGFR was 67 mL/min/1.73 m2, at a median of 7 years post-transplant. The change in eGFR was 2.0 (95% CI 1.0 to 3.1) with dapagliflozin vs 0.3 (95% CI -0.8 to 1.4) with placebo; the difference was 1.7 (95% CI 0.1 to 3.3), p=0.03. There were no unexpected safety signals, and infection rates were similar.

Context: This is the first adequately sized randomized trial of SGLT2 inhibition in heart transplant recipients, and it shows a modest kidney benefit with no excess infections under immunosuppression. That safety finding is arguably as useful as the efficacy result. It extends the kidney-protection logic of the CKD trials to calcineurin-inhibitor nephropathy. Limitations: a 12-month surrogate endpoint, a lower confidence bound near zero, and stable, late recipients only. It reasonably supports adding an SGLT2 inhibitor in stable recipients with declining eGFR, while early post-transplant use remains untested.

Also notable

- Intensive Blood-Pressure Control in Patients with Previous Ischemic Stroke (*N Engl J Med*). OPTIMAL-STROKE: after ischemic stroke or TIA, a systolic target <120 vs <140 mm Hg cut MACE (10.9% vs 13.5%; HR 0.79), with more renal events.
- Discontinuing Statins in Older Adults without Cardiovascular Disease (*N Engl J Med*). STREAM: stopping primary-prevention statins in multimorbid adults ≥70 met noninferiority, yet events were 12.7% vs 9.2%, a weak case for deprescribing.
- Cangrelor versus crushed ticagrelor in acute myocardial infarction complicated by cardiogenic shock (DAPT-SHOCK-AMI): a randomised, double-blind trial (*Lancet*). DAPT-SHOCK-AMI: in infarct-related shock, cangrelor inhibited platelets better than crushed ticagrelor but failed 30-day noninferiority (38% vs 41%).

AI in Medicine

This week's strongest AI work was prospective or randomized, and it was humbling. An AI-ECG-guided AF screening trial found only a fragile gain, and a generative AI pregnancy chatbot did not improve maternal mental health at 12 months. Wearable and EHR tools keep improving at detecting disease or care gaps, but showing that they change outcomes remains the unsolved step.

Leveraging AI-ECG Technology for Early Notification and Tracking of AF Development (LATENT): A Randomized Controlled Trial

Circulation · Lee CC, Lin WY, Liu WT, et al. · Single-center open-label RCT, 14,665 adults · DOI

Area: AI-ECG-guided atrial fibrillation screening

Design: Adults 40-85 without prior AF/flutter were randomized to an AI-ECG strategy, in which a high-risk sinus-rhythm ECG prompted an offer of up to 7 days of Holter monitoring, or usual care with matched controls monitored identically. Primary endpoint: new AF, flutter, or pre-AF atrial findings within 90 days.

Results: AI-ECG flagged 84 (1.1%) intervention and 78 (1.1%) control participants as high risk; 48 (57.1%) were monitored. The primary outcome occurred in 0.74% vs 0.48% (OR 1.54, 95% CI 1.01-2.37; P=0.046), and reclassifying 1 event moved P to 0.057. Yield among high-risk participants was 32.1% vs 2.6% (Pinteraction<0.001). AF/flutter detection (0.3% vs 0.2%) and antiarrhythmic initiation at 180 days (2.3% vs 2.4%) did not differ.

Context: AI-ECG picked out a tiny subgroup with high monitoring yield, but the population-level gain was statistically fragile and driven largely by pre-AF ectopy rather than AF itself. Prior work had shown AI-ECG predicts incident AF; this is one of the first randomized tests of acting on that prediction. Screening trials such as LOOP showed that finding more AF does not reliably reduce stroke, and LATENT lacks any clinical endpoint. Low Holter uptake and a single center limit generalizability. Treat it as feasibility, not justification for deployment.

Opportunistically Detecting Signs of Hypertension on a Consumer Smartwatch

Nature · Di Achille P, Cai L, Wu J, et al. · Prospective multicenter diagnostic validation, 196 participants · DOI

Area: Cuffless hypertension screening, consumer smartwatch

Design: Adults without known cardiovascular disease or antihypertensive therapy wore a consumer smartwatch for seven days. An AI model read photoplethysmography and accelerometry waveforms without cuff calibration and was compared against 24-hour ambulatory BP as reference, with office BP screening as comparator.

Results: The smartwatch algorithm detected hypertension with sensitivity 65.8% (95% CI, 54.0%-76.3%), specificity 90.0% (95% CI, 83.2%-94.7%), and PPV 80.6% (95% CI, 68.6%-89.6%). Office BP screening had sensitivity 55.3% (95% CI, 43.4%-66.7%), the same specificity of 90.0%, and PPV 77.8%.

Context: An uncalibrated smartwatch flagged ambulatory-confirmed hypertension at least as well as an office reading. That supports wearables as a passive prompt toward cuff-based confirmation, not as a BP measurement device. The confidence intervals overlap with office screening, roughly a third of hypertensive participants were missed, and the cohort was small and excluded people with cardiovascular disease. The value is population reach among unaware adults. A negative notification should not reassure.

Generative Artificial Intelligence and Maternal Health Outcomes in Pregnant and Postpartum Individuals: A Randomized Clinical Trial

JAMA Netw Open · Norell EM, Yue X, Mane H, et al. · RCT, 400 participants · DOI

Area: Patient-facing generative AI, maternal mental health

Design: Pregnant or postpartum (infant under 6 months) individuals from racial and ethnic minoritized groups, recruited online across the US, were randomized 1:1 to a generative AI maternal health chatbot or an active control (monthly children's books). Primary outcomes were GAD-7 and PHQ-9 scores at 12 months.

Results: The chatbot did not change anxiety (adjusted mean difference 0.10; 95% CI, -0.77 to 0.96) or depression (-0.23; 95% CI, -1.05 to 0.59). Emergency department visits, postpartum care, contraception use, and sleep duration also did not differ. Among participants pregnant at enrollment, self-reported pregnancy complications were higher with the chatbot in the 12-month analysis but not in longitudinal analyses. In exploratory analyses, higher engagement correlated with longer sleep and higher anxiety.

Context: Simply giving people access to an LLM health chatbot did not improve mental health or care use over a year. This is one of the few randomized, long-follow-up tests of a patient-facing generative AI tool, and it is a clean negative. Most supportive evidence for such tools comes from usability or engagement studies. The intervention was access rather than a structured clinical pathway, which probably limited the dose. Patient-facing LLMs should be held to the same outcome evidence standard as any other intervention.

AI-Identified Reasons for Nonescalation of Lipid-Lowering Therapy for Secondary Prevention

JAMA Cardiol · Retrospective cohort, 28,113 patients · LLM extraction of lipid therapy inertia in ASCVD

An LLM read notes for 13,810 patients with ASCVD and above-goal LDL-C, matching human annotators (F1 0.80-0.98). Therapy was not escalated in 88.7% of relevant notes; the most common documented reason was patient-reported adverse effects (12.6%). The analysis relies on documentation and comes from a single health system.

External Validation of a Machine Learning Model for Delirium Risk

JAMA Netw Open · Quality improvement external deployment, 33,095 admissions · ML delirium risk model, live deployment

A multimodal NLP delirium model deployed without retraining at a second hospital achieved AUROC 0.91, compared with 0.94 at the development site. Monthly delirium detection rose from 9.23% to 19.59%. PPV was only 25.9%, and the pre-post design within one health system limits causal inference.

AI in Medical Education

This week's best work separates AI help in the moment from AI that teaches. A multicenter Doppler trial found AI feedback raised accuracy for novices and experienced clinicians alike, but novices followed wrong advice more often. A yearlong chatbot-simulator RCT showed no learning benefit. A national clerkship survey found most programs lack a clear AI policy and few provide secure tools.

AI feedback for clinician assessment of umbilical and middle cerebral artery Doppler ultrasound: a multicenter randomized trial

Am J Obstet Gynecol MFM · Bashir Z, Wong CK, Lin M, et al. · Multicenter RCT, 96 clinicians · DOI

Area: AI feedback in obstetric Doppler assessment, novices vs experienced

Design: Residents, gynecologists and obstetricians at five Danish hospitals were randomized 1:1 to AI visual and textual feedback or none. Each assessed 40 static umbilical and middle cerebral artery Doppler vignettes (1,920 total) for gate placement or waveform adequacy against expert consensus.

Results: Accuracy was 75.0% with AI feedback vs 62.3% without (mean difference 12.7%; 95% CI 8.2-17.1; $p < 0.001$). Novices improved from 57.9% to 70.0% (difference 12.1%, 95% CI 2.4-21.7) and experienced clinicians from 65.0% to 78.0% (13.0%, 95% CI 5.5-20.5). Novices followed incorrect AI feedback more often (22.3% vs 15.8%; P = .01). AI alone beat unaided novices, but experienced clinicians with AI beat AI alone.

Context: AI feedback raised accuracy by about 13 points regardless of experience, but novices were more likely to follow it when it was wrong. That matches the broader human-AI literature: accuracy gains come with automation bias that falls hardest on trainees, and expertise matters most when the model errs. The tasks were static vignettes, not live scanning, and nobody was tested after the AI was removed. The trial therefore shows assisted performance, not learning, and cannot address deskilling. Supervision models should assume trainees will defer to the machine.

Policy, ethical, and practical considerations of artificial intelligence use in the internal medicine clerkship: results of a survey of clerkship directors

Acad Med · National survey, 114 clerkship directors · AI policy in internal medicine clerkships

At an 80.9% response rate, 39.5% of directors reported no formal AI policy and 34.2% were unsure one existed. Although 73.7% would encourage AI for at least one clinical skill, only 6.1% had firewall-protected tools. The data are self-reported and from 2024.

Evaluating an AI-Powered Patient Simulator for Communication Skills Training: One-Year Randomized Controlled Trial

J Med Internet Res · Nonblinded RCT, 50 medical students · Generative AI patient simulator for communication skills

Over one academic year, UK students practicing with a chatbot simulator showed larger descriptive OSCE communication gains than controls, but the between-group difference was not significant. The trial was single-site and badly underpowered, so a modest effect cannot be excluded.

AI-augmented versus expert-authored multiple-choice questions: a psychometric comparison in a high-stakes specialty examination

BMC Med Educ · Psychometric comparison, 304 examinees · LLM-generated items in high-stakes specialty examinations

ChatGPT-4o-augmented MCQs went into a national specialty exam after multistage expert review. They discriminated comparably and left overall reliability unchanged, but were easier and more often lacked a functioning distractor. Only 12 of 90 generated items reached item analysis.

Longevity Medicine

A thin week in which human data mostly tempered geroprotector enthusiasm. A large transplant registry found no survival signal from chronic mTOR inhibition, and whole-blood NAD tracked cardiovascular disease rather than age, which weakens the premise for routine NAD boosting. The one novel senolytic, an oral mineral supplement acting through a redox-proteostasis vulnerability, remains mouse-only.

Rapamycin and human longevity among kidney transplant recipients surviving the first year

Geroscience · Registry cohort, 11,474 kidney recipients · Rapamycin and human longevity

Propensity-weighted median survival age was 64.6 years with rapamycin (n=4885) and 64.3 years without (n=6589), difference CI -1.6 to 2.0 years, over a median 11 years. Caveat: observational, immunosuppressive dosing in transplant recipients, so it does not test intermittent low-dose geroprotective regimens.

Whole-blood NAD levels track cardiovascular disease but not chronological age

Nat Metab · Cross-sectional cohort, 1,068 adults · NAD as an aging biomarker

Across 1,340 measurements, whole-blood NAD did not fall with age in healthy adults; it was stable in males and rose in females. It was lower in cardiovascular disease and hypertension. Caveat: cross-sectional, so it cannot show that supplementation would help.

Altered protein cysteine reactivity in senescent cells is a vulnerability for senolysis to ameliorate aging

Nat Aging · Mouse and cell mechanistic study · Senolytics, preclinical

Senescent cells accumulate glutathione disulfide via glutathione reductase downregulation, and oral vanadyl sulfate exploits this to cause disulfide-driven protein misfolding and senescent cell death. It improved age-related phenotypes and extended lifespan in male mice. Caveat: preclinical, males only, with no human data.

General Medicine

Two themes stood out this week. First, incretin therapy keeps moving beyond weight loss into specific comorbidities. Retatrutide reduced knee osteoarthritis pain in two phase 3 trials. Oral orforglipron cut apnea-hypopnea index in moderate to severe sleep apnea, with or without PAP. A tirzepatide trial showed that structured resistance training preserves lean mass and strength without blunting fat loss, which

matters for anyone prescribing these drugs to older patients. Second, there is growing evidence for less intensive care in frail older adults. Prepare for Kidney Care, the first randomized comparison of conservative kidney management with dialysis preparation in older frail patients, found no significant difference in quality-adjusted life-years. Outside these themes, a national Epic Cosmos cohort tied rising refusal of newborn vitamin K to more infant bleeding, including roughly 2.4-fold intracranial hemorrhage risk.

Preparing for responsive management versus renal dialysis in older people living with kidney failure and frailty or multimorbidity in the UK (Prepare for Kidney Care): a two-arm, superiority, parallel-group, open-label, pragmatic, phase 3, individually randomised controlled trial

Lancet · Pragmatic open-label RCT, 445 patients · Kidney failure in frail older adults

Older frail or multimorbid patients with kidney failure were randomized to prepare for conservative management or for dialysis; QALYs were similar over 4·5 years (adjusted difference -0·147, p=0·16).

Retatrutide Once Weekly in Adults with Knee Osteoarthritis Pain and Obesity

N Engl J Med · Two phase 3 RCTs, 445 and 574 patients · Knee osteoarthritis and obesity

In TRIUMPH-4, retatrutide reduced WOMAC pain by -3.96 (9 mg) and -3.74 (12 mg) vs -2.12 with placebo at 68 weeks, with up to -23.7% weight loss.

Orforglipron for Obstructive Sleep Apnea: Results From 2 Randomized Clinical Trials (ATTAIN-OSA)

JAMA · Two phase 3 RCTs, 363 and 349 patients · Obstructive sleep apnea and obesity

Oral orforglipron reduced AHI by -17.8 vs -4.5 events/h without PAP and -17.3 vs -5.3 with PAP at 52 weeks; GI events 68.0% vs 37.4%.

Resistance exercise to attenuate lean mass loss during tirzepatide therapy for obesity: a randomized controlled trial

Nat Med · RCT, 120 patients · Obesity pharmacotherapy and lean mass

Adding thrice-weekly resistance training to tirzepatide reduced lean mass loss (-3.57 vs -4.67 kg; difference 1.09 kg) and increased strength, with similar weight loss.

Nonreceipt of Newborn Vitamin K and Risk of Bleeding

JAMA · Retrospective national cohort (Epic Cosmos) · Newborn vitamin K prophylaxis

Vitamin K nonreceipt rose from 3.1% to 6.8% (2017-2025). Infants who did not receive it had higher intracranial bleeding risk (ARR 2.43; 95% CI, 2.26-2.62).

Verification Notes

- 497 candidate papers screened from PubMed; 22 selected (9 with a full write-up).
- Every selected PMID, title, author list and journal is taken directly from PubMed. Design and Results lines are drawn from the abstract and their numbers checked against it automatically. Context lines reflect the model's knowledge of the prior literature and are not machine-verified.
- No discrepancies flagged.

Selected and written by claude-opus-5-5 from PubMed records added Nov 04 – Nov 11, 2026.